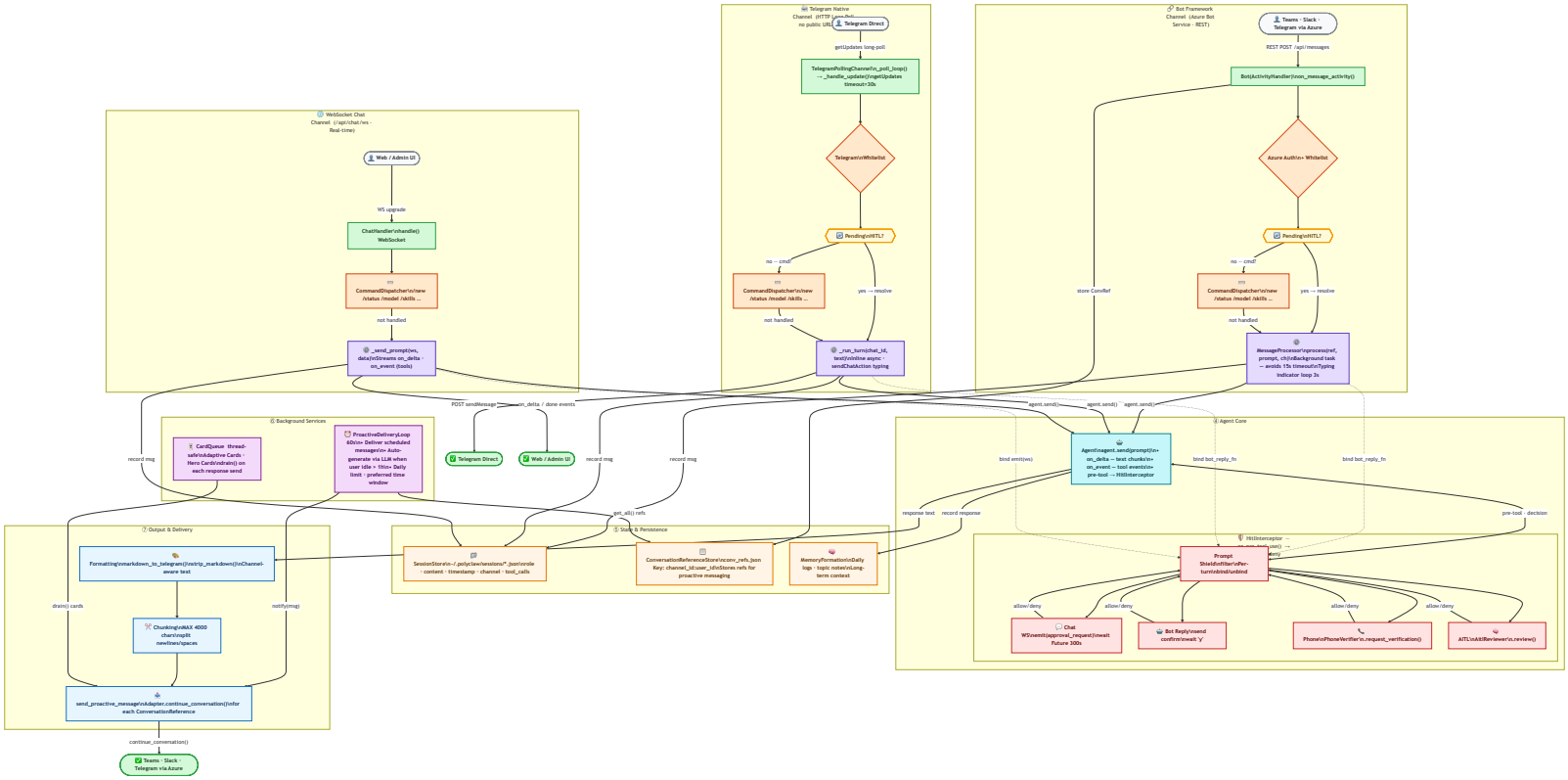
Telegram Native Channel (HTTP polling — no public URL)
Telegram Direct
getUpdates long-poll
TelegramPollingChannel\n_poll_loop() → _handle_update()\ngetUpdates timeout=30s
Telegram\nWhitelist
Pending\nHITL?
no → cmd?
CommandDispatcher\n/new /status /model /skills …
yes → resolve
not handled
_run_turn(chat_id, text)\nInline async : sendChatAction typing
Bot Framework Channel (Azure Bot Service · REST)
Teams · Slack · Telegram via Azure
REST POST /api/messages
Bot(ActivityHandler)\non_message_activity()
Azure Auth\n+ Whitelist
Pending\nHITL?
no → cmd?
CommandDispatcher\n/new /status /model /skills …
yes → resolve
not handled
MessageProcessor\nprocess(ref, prompt, ch)\nBackground task — avoids 15s timeout\nTyping indicator loop 3s
store ConvRef
WebSocket Chat Channel (/api/chat/ws · Real-time)
Web / Admin UI
WS upgrade
ChatHandler\nhandle() WebSocket
CommandDispatcher\n/new /status /model /skills …
not handled
_send_prompt(ws, data)\nStreams on_delta · on_event (tools)
POST sendMessage
on_delta / done events
agent.send()
agent.send()
agent.send()
⑥ Background Services
CardQueue thread-safe\nAdaptive Cards · Hero Cards\ndrain() on each response send
ProactiveDeliveryLoop 60s\n• Deliver scheduled messages\n• Auto-generate via LLM when user idle > 1hto• Daily limit · preferred time window
Telegram Direct
Web / Admin UI
record msg
record msg
record msg
④ Agent Core
Agent\nagent.send(prompt)\n• on_delta — text chunks\n• on_event — tool events\n• pre-tool — HitlInterceptor
bind emit(ws)
bind bot_reply_fn
bind bot_reply_fn
response text
record response
pre-tool · decision
HitlInterceptor — intercept() — every
Prompt Shield\nfilter\nPer-turn\nbind\nunbind
allow / deny
allow / deny
allow / deny
allow / deny
Chat WS\nemit(approval_request)\nwait Future 300s
Bot Reply\nsend confirm\nwait 'y'
Phone\nPhoneVerifier\n.request_verification()
AITL\nAITLReviewer\n.review()
set_all() refs
⑤ State & Persistence
SessionStore\n~/.polyclaw/sessions/*.jsonl\nrole · content · timestamp · channel · tool_calls
ConversationReferenceStore\nconv_refs.json\nKey: channel_id:user_id\nStores refs for proactive messaging
MemoryFormation\nDaily logs · topic notes\nLong-term context
⑦ Output & Delivery
Formatting\nmarkdown_to_telegram()\nstrip_markdown()\nChannel-aware text
Chunking\nMAX 4000 chars\nsplit newlines/spaces
send_proactive_message\nAdapter.continue_conversation()\nfor each ConversationReference
drain() cards
notify(msg)
continue_conversation()
Teams · Slack · Telegram via Azure